16 y is inversely proportional to the square of x.
$y = 1$ when $x = 10$

Find the value of y when $x = 5$

$$y = \frac{k}{x^2}$$

1 mark
Inverse-square relationship.

$$1 = \frac{k}{10^2}$$

$k = 100$

1 mark
Find constant k.

When $x = 5$, $y = \frac{100}{5^2} = 4$

1 mark
Substitute x=5.

$y =$ .....

(Total for Question 16 is 3 marks)